

Create and use algorithms involving a sequence of steps and decisions and digital tools to experiment with factors, multiples and divisibility; identify, interpret and describe emerging patterns

Learning intention

- create and use algorithms involving a sequence of steps and decisions and digital tools to experiment with factors, multiples and divisibility; identify, interpret and describe emerging patterns.

Teach and model

- identifying lowest common multiples and highest common factors of pairs or triples of natural numbers.
- creating algorithms that use multiplication and division facts to determine if a number is a multiple or factor of another number.
- using the “fill down” function of a spreadsheet and a multiplication formula to generate a sequence of numbers that represent the multiples of any number you enter into the cell; describing and explaining the emerging patterns.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

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Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.